

Assignment: DP for Constrained Sustainable Manufacturing

Industrial Systems Optimization

Academic Year 2026

Problem Overview

You are tasked with scheduling n jobs on a single high-precision 3D Metal Printer. To minimize the **Carbon Footprint**, you must find a sequence that minimizes the total energy consumed during **Atmospheric Resets** (setups) while strictly adhering to **Precedence Constraints**.

Operational Context: Aerospace Component Printing

The 3D printer uses different metal alloys (Titanium, Aluminum, Inconel).

- **Sequence-Dependent Energy** (e_{ij}): Changing from one alloy to another requires purging the chamber and resetting the inert gas atmosphere. The energy cost depends on the chemical compatibility of the two alloys.
- **Precedence Constraints** ($j \rightarrow k$): Due to the assembly logic of modular aerospace parts, certain components (e.g., the "Core Chassis") *must* be completed before the auxiliary "Brackets" can start, as they share the same build plate coordinates in a nested sequence.

Dynamic Programming Formulation

To solve this via DP, define the state (S, i) as the minimum setup energy to complete the set of jobs S , ending with job i .

The Recursive Step

For a set S and a new job $j \notin S$, the transition to state $(S \cup \{j\}, j)$ is valid **if and only if**:

$$\text{Predecessors}(j) \subseteq S$$

The value is calculated as:

$$V(S \cup \{j\}, j) = \min_{i \in S} \{V(S, i) + e_{ij}\}$$

Deterministic Problem Instance

Table 1: Job List and Technical Dependencies

Job ID	Alloy Type	Processing Time	Must follow...
Job_1	Titanium (T)	120 min	None
Job_2	Aluminum (A)	90 min	Job_1
Job_3	Titanium (T)	110 min	None
Job_4	Inconel (I)	150 min	Job_2 and Job_3

Table 2: Alloy Reset Energy (e_{ij} in kWh)

From ↓ / To →	Job 1 (T)	Job 2 (A)	Job 3 (T)	Job 4 (I)
Start	10	15	10	20
Job 1 (T)	-	40	5	30
Job 2 (A)	35	-	35	25
Job 3 (T)	5	40	-	30
Job 4 (I)	20	20	20	-

Assignment Task

Implement the DP algorithm. Your code must:

1. Check the precedence feasibility before exploring a state.
2. Output the sequence that minimizes the total "Atmospheric Reset" energy.
3. Identify how many "impossible" permutations were pruned by the precedence constraints.